MSD MANUAL
Professional Version

COVID-19

(SARS-CoV-2 Infection)

*By **Sophie Katz**, MD, MPH, Vanderbilt University Medical Center*
*Reviewed By **Brenda L. Tesini**, MD, University of Rochester School of Medicine and Dentistry*
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COVID-19 is a respiratory illness caused by the coronavirus SARS-CoV-2. Infection may be asymptomatic or have symptoms ranging from mild upper respiratory symptoms to acute respiratory failure and death. COVID-19 can involve multiple organ systems (eg, cardiac, renal, neurologic, coagulation). Prevention is by vaccination and infection control precautions (eg, face masks, handwashing, social distancing, isolation of infected individuals). Diagnosis is by antigen or PCR (polymerase chain reaction) testing of upper or lower respiratory secretions. Treatment is with supportive care, antiviral medications, or corticosteroids.

SARS-CoV-2

3D MODEL

COVID-19 was first reported in late 2019 in Wuhan, China and declared a pandemic by the World Health Organization (WHO) in March 2020. It is caused by SARS-CoV-2, a coronavirus discovered in 2019. SARS-CoV-2 infection causes a spectrum of severity of disease, from asymptomatic to acute respiratory failure and death. Risk factors for severe disease include older age, immunocompromise, comorbidities (eg, diabetes, chronic kidney disease), and pregnancy. Vaccines have shown to be somewhat effective in preventing transmission and very effective in preventing severe disease and mortality.

For current information on the number of cases and fatalities, see the Centers for Disease Control and Prevention (CDC): COVID Data Tracker and the WHO Coronavirus (COVID-19) Dashboard.

Transmission of COVID-19

SARS-CoV-2 spreads by close person-to-person contact, mainly via respiratory droplets produced when an infected person coughs, sneezes, sings, exercises, or talks. The spread occurs through large respiratory droplets that can travel short distances and land directly on mucosal surfaces or through small respiratory particle aerosols that can linger in air for several hours and travel longer distances (approximately > 6 feet [2 meters]) before being inhaled. Spread of the virus could also occur via contact with surfaces contaminated (fomites) by respiratory secretions, if a person touches a contaminated surface and then touches a mucous membrane on the face (eyes, nose, mouth).

SARS-CoV-2 spreads easily between people. The risk of transmission is directly related to the amount of virus to which a person is exposed. In general, the closer and longer the interaction with an infected person, the higher the risk of virus spread. Both asymptomatic and symptomatic patients can transmit the virus, making it difficult to control spread. A symptomatic person is most contagious for the several days before and after the onset of symptoms, at which time the viral load in respiratory secretions is greatest.

Factors such as distance from an infected person, the number of infected people in the room, the duration of time spent with infected people, the size of the air space, aerosol-generating activity (eg, singing, shouting, or exercising), ventilation in the location, and the direction and speed of airflow can contribute to this risk.

Genetic variants of SARS-CoV-2 emerge as it evolves. Variants with the potential for increased transmissibility, more severe disease, diagnostic detection failures, or reduced response to available treatments and/or vaccines are tracked as Variants of Concern and are commonly referred to by their WHO-designated Greek alphabet label or their Pango lineage number. A genetic mutation that confers a fitness advantage, namely increased transmissibility, can rapidly replace previously circulating variants. The progression of dominant variants in the United States and much of the world includes Alpha, Beta, Delta, and Omicron. The Omicron variant has predominated worldwide since March 2022, with newer and more transmissible Omicron subvariants replacing the original Omicron. See also CDC: Variants & Genomic Surveillance.

Situations with high risk of transmission include congregate living facilities (eg, elder care or other long-term care facilities, residential schools, prisons, ships) as well as crowded, poorly ventilated environments, such as indoor religious services, gyms, bars, nightclubs, indoor restaurants, and meat-packing facilities. Such situations have a high population density in which maintaining distance and ventilation precautions is difficult. The residents of elder care facilities are also at high risk of severe disease because of age and underlying medical disorders. Large indoor events or private gatherings such as meetings or weddings have also been associated with high transmission rates. These so-called super-spreader events or situations are probably due to a combination of biological, environmental, and behavioral factors.

Social determinants of health (conditions in the places where people are born, live, learn, work, and play) impact a wide range of health risks and outcomes, such as exposure to SARS-CoV-2 infection,

severe COVID-19, and death, as well as access to testing, vaccination, and treatment. In the United States, COVID-19 case, hospitalization, and death rates were higher in some racial and ethnic minority groups, including among people who are Black, Hispanic or Latino, American Indian, and Alaska Native, but have become similar in all ethnic and minority groups since April 2024 (CDC: COVID Data Tracker).

For people infected with COVID-19, isolation and precaution measures are recommended in an attempt to limit the spread of SARS-CoV-2 infection. (See also CDC: Preventing Spread of Respiratory Viruses When You're Sick.) For people exposed to COVID-19, precautions are recommended to help reduce the risk of spreading the virus. (See also CDC: Preventing Respiratory Viruses.)

Symptoms and Signs of COVID-19

The severity and constellation of symptoms vary in people with COVID-19. Some have few to no symptoms, and some become severely ill and die. Symptoms may include

- Fever
- Cough
- Sore throat
- Congestion or runny nose
- Shortness of breath or difficulty breathing
- Chills or repeated shaking with chills
- New loss of smell or taste
- Fatigue
- Muscle pain
- Headache
- Nausea or vomiting
- Diarrhea

The incubation period (ie, time from exposure to symptom onset) ranges from 2 to 10 days, with a median estimated to be only 2 to 4 days for the Omicron variant (1). Many infected people have no symptoms or mild disease; the likelihood of this varies depending on the SARS-CoV-2 variant and person's risk for severe disease, including COVID immunization status.

Severe disease is characterized by dyspnea, hypoxia, and extensive lung involvement on imaging. This can progress to respiratory failure requiring mechanical ventilation, shock, multiorgan failure, and death.

Risk factors for severe disease

The risk of serious disease and death in COVID-19 cases increases in people over age 65, in people who smoke cigarettes or previously smoked, and in people with other serious medical disorders, such as

- Cancer

- Chronic heart, lung, kidney, or liver disease
- Cystic fibrosis
- Diabetes
- Stroke or cerebrovascular disease
- Immunocompromising conditions
- HIV infection
- Tuberculosis
- Sickle cell disease
- Thalassemia
- Dementia
- Obesity
- Pregnancy (up to 42 days after pregnancy)
- Some types of disabilities
- Substance use disorders
- Physical inactivity
- Some mental health disorders such as depression and schizophrenia

(Also see CDC: COVID-19: People with Certain Medical Conditions.)

Vaccination dramatically lowers the risk of severe illness for all age groups—lower vaccination rates in younger age groups has shifted the age demographic of hospitalized patients (see CDC: COVID Data Tracker). People who have been vaccinated against COVID-19 can be asymptomatic or have fewer and less severe symptoms that resolve at a much faster rate compared to unvaccinated people.

Complications

In addition to respiratory disease that can progress to acute respiratory distress syndrome (ARDS) and death, other serious complications include the following:

- Heart disorders, including arrhythmias, cardiomyopathy, and acute cardiac injury
- Coagulation disorders, including thromboembolism and pulmonary emboli, disseminated intravascular coagulation (DIC), hemorrhage, and arterial clot formation
- Guillain-Barré syndrome (rare)
- Sepsis, shock, and multiorgan failure

A postinfectious inflammatory syndrome termed **multisystem inflammatory syndrome in children (MIS-C)** has been observed as a rare complication of SARS-CoV-2 infection. It has features similar to Kawasaki disease or toxic shock syndrome. Children with MIS-C most commonly present with fever, tachycardia, signs of systemic inflammation, and multisystem involvement (eg, cardiac, gastrointestinal, renal) at 2 to 6 months following a generally mild or even asymptomatic SARS-CoV-2 infection. Cases meeting the following criteria should be reported to local, state, or territorial health departments as suspected MIS-C: individuals < 21 years old with fever > 24 hours, laboratory evidence of inflammation,

signs of severe multisystem (≥ 2 organs) involvement requiring hospitalization, and laboratory or epidemiologic association with recent SARS-CoV-2 infection (2). Vaccination appears to be highly protective against the development of MIS-C (2). MIS-C has become less prevalent since the start of the pandemic, likely due to vaccinations and immunity following prior SARS-CoV-2 infection. A similar multisystem inflammatory syndrome in young and middle-aged adults (MIS-A) also has been reported (3).

Symptom resolution

In most patients, symptoms resolve within about a week. However, some patients begin with mild symptoms, then clinically deteriorate after a week, progressing to severe disease, including ARDS. Prolonged illness appears to be more common in those with severe disease, but even patients with mild illness may have persistent symptoms, including dyspnea, cough, and malaise that last for weeks or even months. Viral PCR tests in patients may remain positive for at least 3 months regardless of symptoms. However, even patients with lingering symptoms are generally not considered infectious, as virus is rarely able to be cultured from the upper respiratory tract of patients after 10 days of illness.

COVID-19 may also be associated with long-term sequelae following acute illness (4), and symptoms from post–COVID-19 conditions can linger for months. This has been referred to by many names, including long COVID, long-haul COVID, and post-acute COVID-19 syndrome or condition, and is estimated to impact 25 to 50% of all patients in some United States surveys. Fatigue, weakness, pain, myalgias, dyspnea, and cognitive dysfunction are commonly reported and can be associated with a poorer quality of life (5). Risk factors for long-term sequelae may include more severe disease presentation, older age, female sex, and pre-existing lung disease. An international case definition has been established to aid in the diagnosis and further investigation of this condition (6).

Symptoms and signs references

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Diagnosis of COVID-19

- Real-time reverse transcriptase–polymerase chain reaction (RT-PCR) or other nucleic acid amplification test (NAAT) of upper and lower respiratory secretions
- Antigen testing of upper respiratory secretions

Diagnostic testing for COVID-19 is available through laboratories and public testing sites and can also be done at home. There are 2 main types of diagnostic COVID-19 tests: real-time reverse transcriptase–polymerase chain reaction (RT-PCR) (or other nucleic acid amplification test [NAAT]) and antigen tests. The choice of diagnostic test and its interpretation should be influenced by the likelihood of the person having COVID-19 based on the prevalence of SARS-CoV-2 in the population and the presence of COVID-19 symptoms, signs, or close contact with a known case of COVID-19 (see CDC: COVID-19: Overview of Testing for SARS-CoV-2, the virus that causes COVID-19).

RT-PCR has the highest analytical sensitivity and specificity and is the gold standard diagnostic test for COVID-19. Other NAAT platforms are generally slightly less sensitive than RT-PCR with equivalent specificity (see CDC: COVID-19: Nucleic Acid Amplification Tests). Positive viral PCR tests, however, do not always indicate active infection. They can detect nonviable viral nucleic acid fragments and may remain positive for at least 3 months after initial diagnosis regardless of symptoms.

Point-of-care and home-based antigen testing can provide rapid results (see CDC: COVID-19: Considerations for SARS-CoV-2 Antigen Testing for Healthcare Providers Testing Individuals in the Community). This can be an important measure to identify asymptomatic cases and interrupt SARS-CoV-2 transmission. Point-of-care or home-based antigen detection tests are less sensitive than NAATs, particularly at the onset of infection when viral load may be lower. The sensitivity of these tests compared to PCR tests vary by manufacturer and time course of infection, with ranges reported from 40 to 90% (1-4). Therefore, it may be necessary to confirm some antigen test results (eg, a negative test in a person with symptoms) with an RT-PCR or other NAAT. Many antigen-detection test kits also recommend repeating the test serially over several days to increase the likelihood of detecting infection (see also FDA: At-Home COVID-19 Antigen Tests-Take Steps to Reduce Your Risk of False Negative Results). Some tests may not detect the Omicron variant or other emerging variants (see FDA: SARS-CoV-2 Viral Mutations: Impact on COVID-19 Tests). Antigen tests are less likely to stay positive following resolution of infection since they only detect higher viral loads. However, other factors in addition to viral load can influence infectivity; therefore antigen test results do not necessarily correlate with infectiousness.

Acceptable specimens for COVID-19 diagnostic testing include nasopharyngeal, oropharyngeal, nasal mid-turbinate, anterior nares, and saliva. Refer to the accepting laboratory's collection instructions or test kit package insert instructions, because not all testing platforms and laboratories may be able to test all specimen types. These may be collected by a health care practitioner or self-collected, with the exception of nasopharyngeal specimens, which should only be collected by an appropriately trained and credentialed health care practitioner.

For nasopharyngeal and oropharyngeal specimens, use only synthetic fiber swabs with plastic or wire shafts. Do not use calcium alginate swabs or swabs with wooden shafts, as they may contain substances

that inactivate some viruses and inhibit PCR testing. The swabs should be placed immediately into the transport tube provided. Maintain proper infection control when collecting specimens.

For biosafety reasons, local institutions and laboratories should not attempt to isolate the virus in cell culture.

Serologic, or antibody, testing should not be used to diagnose acute COVID-19 illness, because antibodies most commonly become detectable only 1 to 3 weeks after symptom onset. Antibody tests are available that target the SARS-CoV-2 nucleocapsid antigen, spike antigen, and the receptor binding domain of the spike antigen. Tests that detect antibody to the nucleocapsid protein are recommended to evaluate for evidence of prior infection in vaccinated persons as that antigen is not included in the vaccine. Quantitative and semi-quantitative antibody assays are available, but there is currently no accepted correlate of immunity, and testing is not recommended to determine immune response to vaccination or infection (see Infectious Disease Society of America (IDSA): Guidelines on the Diagnosis of COVID-19: Serologic Testing, Executive Summary).

Evaluating symptomatic patients

Routine laboratory findings for those with more severe disease include lymphopenia as well as less specific findings of elevated aminotransaminase (ALT, AST) levels, elevated lactate dehydrogenase (LDH) levels, D-dimer, ferritin, and elevated inflammatory markers such as C-reactive protein.

Patients with dyspnea, hypoxia on home oximetry, or other concerning symptoms should be referred for in-person medical evaluation, including oxygen saturation measurement, and followed up with for signs of clinical deterioration.

Chest imaging findings can be normal with mild disease and increase with increasing severity of the illness. Typical findings are consistent with viral pneumonia and include ground-glass opacities and consolidation on either chest x-ray or chest CT. Chest imaging is not recommended as a routine screening tool for COVID-19.

Diagnosis references

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Treatment of COVID-19

- Supportive care
- Sometimes, for mild to moderate illness with high-risk of severe disease: nirmatrelvir/ritonavir combination; remdesivir (short course); molnupiravir
- For severe illness: remdesivir; dexamethasone; immunomodulators

Treatment of COVID-19 depends on the severity of illness and the likelihood that the patient will develop severe disease. Treatment guidelines are evolving as new data emerge (see Infectious Diseases Society of America (IDSA) Guidelines on the Treatment and Management of Patients with COVID-19 and World Health Organization (WHO): COVID-19 Treatments).

The IDSA definitions of severity to target treatments are as follows:

- Mild to moderate COVID-19 (SpO2 > 94% on room air and not needing supplemental oxygen) with risk factors for progression to severe disease, hospitalization, or death
- Severe but not critical COVID-19 (SpO2 < 94% on room air or needing low-flow supplemental oxygen)
- Critical COVID-19 needing high-flow oxygen or noninvasive ventilation
- Critical COVID-19 needing mechanical ventilation or extracorporeal membrane oxygenation (ECMO)

The effectiveness of particular antiviral medications and monoclonal antibodies against locally circulating variants is considered when making treatment decisions (see Open Data Portal: Database of in vitro therapeutic activity against SARS-CoV-2 variants). Treatment options are listed in order of preference based on currently available data and circulating SARS-CoV-2 variants. Treatment choice should be based on medication availability, infrastructure to administer the medication, and patient-specific factors that include symptom duration, potential drug interactions, and liver and renal impairment. There are no data regarding combination treatments with the currently available therapies; therefore, only one anti-SARS-CoV-2 medication should be administered.

Early treatment for patients with mild to moderate COVID-19 who are high risk for progression to severe illness

These treatments are intended to prevent progression to severe disease in high-risk patients. They are given to patients within a few days of onset of mild to moderate COVID-19 who are ambulatory (or hospitalized for reasons other than COVID-19); treatment options (other than remdesivir) have not been studied in patients hospitalized for COVID-19.

Nirmatrelvir, an oral antiviral medication, given in combination with **ritonavir,** received approval from the US Food and Drug Administration (FDA) for treatment of mild to moderate COVID-19 in adults in May 2023 and remains under Emergency Use Authorization (EUA) for adolescents (≥ 12 years of age weighing ≥ 40 kilograms) who have positive results of direct SARS-CoV-2 viral testing and who are at high risk for progression to severe COVID-19. The nirmatrelvir/ritonavir combination should be initiated as soon as possible after diagnosis of COVID-19 and within 5 days of symptom onset. The dosage is two

150-mg tablets of nirmatrelvir and one 100-mg tablet of ritonavir taken together orally twice a day for 5 days. This treatment is not authorized for use for longer than 5 consecutive days. (See also the FDA EUA Factsheet.)

Nirmatrelvir is a protease inhibitor that cleaves a SARS-CoV-2 protein and therefore inhibits the virus from replicating. Ritonavir is a strong cytochrome P450 (CYP) 3A4 inhibitor and acts as a boosting agent; it slows down the metabolism of nirmatrelvir, causing it to remain in the body for a longer duration at higher concentrations.

The nirmatrelvir/ritonavir combination was studied in a randomized controlled trial of 2,246 nonhospitalized symptomatic adults ≥ 18 years of age with a prespecified risk factor for progression to severe disease or ≥ 60 years of age regardless of prespecified chronic medical conditions. All patients had not received a COVID-19 vaccine, had not been previously infected with COVID-19, and had a laboratory confirmed diagnosis of SARS-CoV-2 infection. The nirmatrelvir/ritonavir combination given at ≤ 5 days after symptom onset reduced the proportion of people with COVID-19-related hospitalization or death from any cause through day 28 compared to placebo by 88% (0.8 versus 6.3%) (1).

Using the nirmatrelvir/ritonavir combination in people with uncontrolled or undiagnosed HIV-1 infection may lead to HIV-1 drug resistance. The nirmatrelvir/ritonavir combination may cause liver damage, so caution should be exercised in patients with preexisting liver disease, liver enzyme abnormalities, or hepatitis. The medication is not recommended in patients with severe liver disease.

In patients with moderate renal impairment (an estimated creatinine clearance of 30 to 60 mL/minute), the dose is reduced to 150 mg nirmatrelvir (one 150-mg tablet) with 100 mg ritonavir (one 100-mg tablet), with both tablets taken together twice a day for 5 days. Nirmatrelvir/ritonavir combination is not recommended in patients with severe kidney impairment (eGFR < 30 mL/minute).

Return of symptoms has been reported in some patients after use of nirmatrelvir/ritonavir, and PCR and antigen tests for SARS-CoV-2 can become positive again, even in patients who remain asymptomatic. Additional treatment is not currently recommended, but patients should be advised to re-isolate if rebound symptoms or tests occur. (See also CDC: COVID-19 Rebound After Paxlovid Treatment.)

The nirmatrelvir/ritonavir combination medication has a variety of serious known and possible drug interactions; concomitant drugs must be screened for these prior to initiation of treatment. For a list of these drug interactions see the FDA EUA Fact Sheet.

Remdesivir, an intravenous antiviral medication, is used to treat mild to moderate COVID-19 in adults and adolescents (≥ 12 years old and weighing ≥ 40 kilograms) who have positive results of direct SARS-CoV-2 viral testing and who are at high risk for progression to severe COVID-19. It should be initiated as soon as possible and within 7 days of symptom onset. A 3-day course of remdesivir is given, which can be extended to 5 days for patients who progress to severe disease.

Remdesivir was studied for this indication in a randomized controlled trial of 562 nonhospitalized patients age ≥ 12 years of age with symptomatic COVID-19 at high risk for progression to severe disease and who had not received a COVID-19 vaccine. Remdesivir given for 3 days and started within 7 days of

symptom onset reduced hospitalization or death rates through day 28 compared to placebo by 87% (0.7 versus 5.3%) (2).

Molnupiravir, an oral antiviral medication, is a nucleoside analogue that works by introducing errors into the SARS-CoV-2 viral genome, which inhibits viral replication. It has received FDA EUA for treatment of mild to moderate COVID-19 in nonhospitalized adults ≥ 18 years of age who have positive results of direct SARS-CoV-2 viral testing and who are at high risk for progression to severe disease, including hospitalization or death, and for whom alternative COVID-19 treatment options authorized by the FDA are not accessible or clinically appropriate. Molnupiravir should be initiated as soon as possible after diagnosis of COVID-19 and within 5 days of symptom onset. Molnupiravir is administered as four 200-mg capsules taken orally every 12 hours for 5 days. It is not authorized for use for longer than 5 consecutive days. (See also the FDA EUA Fact Sheet for Molnupiravir.)

Molnupiravir may have an effect on the emergence of new SARS-CoV-2 variants based on a theoretical concern; however, the risk is believed to be low based on available genotoxicity data and the limited 5-day treatment course. Molnupiravir is not authorized for use in patients < 18 years of age because it may affect bone and cartilage growth. Molnupiravir is not recommended for use during pregnancy because animal reproduction studies suggested that molnupiravir may cause fetal harm when administered to pregnant patients. Females of childbearing potential are advised to use a reliable method of birth control correctly and consistently during treatment with molnupiravir and for 4 days after the final dose. Males of reproductive potential who are sexually active with females of childbearing potential are advised to use a reliable method of birth control correctly and consistently during treatment with molnupiravir and for at least 3 months after the final dose.

Molnupiravir was studied in a randomized placebo-controlled trial of 1433 nonhospitalized patients ≥ 18 years of age with mild to moderate COVID-19 at high risk for progression to severe COVID-19 and/or hospitalization and who had not received a COVID-19 vaccine. Molnupiravir given for 5 days and started within 5 days of symptom onset reduced by 30% hospitalization or death rates through day 29 compared to placebo (6.8 versus 9.7%) (3).

Bamlanivimab/etesevimab, casirivimab/imdevimab, sotrovimab, and bebtelovimab are neutralizing anti-SARS-CoV-2 monoclonal antibody (mAb) therapies. The FDA currently recommends against their use in treatment of COVID-19 because the dominant Omicron subvariants are resistant to their neutralizing activity.

Treatment for patients with severe COVID-19

Recommended treatment options for severe infection include the antiviral medication remdesivir, the corticosteroid dexamethasone, and additional immunomodulatory medications such as baricitinib, tocilizumab, and sarilumab. These may be used in combination, and treatment decisions should take into account the patient's phase of illness, often characterized by the degree of hypoxia and respiratory support.

Antiviral medications are more likely to provide benefit earlier in the course when illness is a result of active viral replication, whereas anti-inflammatory and immunomodulatory therapies are better suited

for later in the course when the host inflammatory response and immune dysregulation are driving the disease state. (Also see IDSA Guidelines on the Treatment and Management of Patients with COVID-19.)

For patients requiring supplemental oxygen but not additional respiratory support, treatment options include

- Remdesivir alone
- Dexamethasone alone
- Remdesivir plus dexamethasone

The antiviral medication **remdesivir** is used to treat adult and pediatric patients ≥ 28 days of age and weighing ≥ 3 kg who require hospitalization for COVID-19. The recommended treatment duration is 5 days but is approved for use up to 10 days in patients who require invasive mechanical ventilation and/or extracorporeal membrane oxygenation (ECMO). Benefits of antiviral treatment in this group though are inconclusive.

In a large randomized clinical trial (ACTT-1), remdesivir was associated with earlier clinical improvement for patients requiring supplemental oxygen but not mechanical ventilation or other higher levels of support (4). This improved time to recovery was also observed in an outpatient randomized controlled trial (PINETREE) (2). However, benefit was not observed in 2 open-label trials (Solidarity [5] and DisCoVeRy [6]). The World Health Organization (WHO) issued a conditional recommendation against the use of remdesivir for hospitalized patients citing no evidence to suggest that it is an effective treatment for COVID-19 (see WHO: COVID-19 Treatments).

Overall, studies of remdesivir support its use early in infection (prior to days 7 to 10) when active viral replication is more likely to be contributing to illness. Remdesivir can also be considered in patients who are hospitalized but not requiring supplemental oxygen, although data are lacking in this population. Remdesivir is not recommended for patients with an eGFR < 30 mL/minute. Renal function should be monitored before and during remdesivir treatment.

The corticosteroid **dexamethasone** is generally recommended in patients with COVID-19 who require supplemental oxygen, but its use is not recommended in patients who do not require supplemental oxygen. Dexamethasone showed a survival benefit for those requiring supplemental oxygen or mechanical ventilation in the RECOVERY trial (7). Its benefit is likely to be greatest in patients whose illness is due to the inflammatory response from infection. If dexamethasone is not available, other corticosteroids (eg, prednisone, methylprednisolone, hydrocortisone) may be used.

The combination of remdesivir and dexamethasone is commonly used in hospitalized patients requiring supplemental oxygen within the first 10 days of illness when both viral replication and host inflammation may be contributing to the clinical presentation.

For patients requiring noninvasive ventilation (including high flow oxygen delivery systems)

- Dexamethasone is recommended for all patients.
- Remdesivir may be added with particular consideration for patients within 7 to 10 days of symptom onset.

- Additional immunomodulatory medications should be considered, particularly for patients with rapid deterioration or signs of systemic inflammation.

Additional immunomodulators include the JAK inhibitor **baricitinib** (or tofacitinib if unavailable) or the IL-6 inhibitor **tocilizumab** (or sarilumab if unavailable). These recommendations are based on subgroup analyses of numerous randomized (COV-BARRIER [8], ACTT-2 [9]) and open label (REMAP-CAP [10], RECOVERY [11]) clinical trials that showed survival benefit with addition of one of these medications in patients requiring this level of respiratory support. These medications are potent immunosuppressants, and the potential benefit should be weighed against the additional immunosuppressive risk in patients with suspicion of a concomitant serious bacterial or fungal infection or at high risk for opportunistic infections due to an underlying immunosuppressive condition.

For patients requiring mechanical ventilation or ECMO, dexamethasone is recommended for all patients. The addition of tociluzumab should be considered for patients within 24 hours of admission to the intensive care unit (ICU).

Many therapies have been considered and *are not currently recommended* for the treatment or prevention of COVID-19:

- Vilobelimab is an anti-complement (anti-C5a) monoclonal antibody that received an FDA EUA in April 2023 for treatment of COVID-19 in hospitalized adults when initiated within 48 hours of receiving invasive mechanical ventilation or extracorporeal membrane oxygenation. Data about this agent are limited; in a single randomized trial, the presence of a significant effect on 28-day mortality varied by analysis (12). Clinical recommendations for its use have not been published.
- Convalescent plasma is not currently recommended for the treatment of patients hospitalized with COVID-19. Data from a number of randomized clinical trials (13) and a large registry associated with an expanded access program failed to show a meaningful benefit in this population and suggest a potential association with increased need for mechanical ventilation. It may be considered in nonhospitalized patients with mild to moderate disease and high risk of progression to severe disease if no other treatment options are available.
- Nonspecific immunoglobulin (IVIG) and mesenchymal stem cell therapy are also not recommended.
- Additional immunomodulatory therapies, including interferons, kinase inhibitors, and interleukin inhibitors have been used, but there are insufficient data to recommend their routine use outside of clinical trials.
- Other medications that have been used include azithromycin and antiretrovirals, but there are insufficient data to support the use of these medications outside of clinical trials.
- Multiple clinical trials of the HIV retroviral lopinavir/ritonavir and the antimalarial medications chloroquine and hydroxychloroquine have shown these medications to be without benefit (14, 15). The combined lack of benefit and toxicities associated with chloroquine and hydroxychloroquine led to recommendations that they not be used for treatment of COVID-19.

- There are also no randomized clinical trials documenting the usefulness of the antiparasitic medication ivermectin for the prevention or treatment of COVID-19 (16). A recent large randomized, placebo-controlled study showed no effect of ivermectin (17). The FDA and other organizations have issued warnings about toxicity from the inappropriate use of ivermectin preparations intended for large animal use (see FDA: Ivermectin and COVID-19).
- The selective serotonin reuptake inhibitor (SSRI) fluvoxamine has been studied in 2 randomized clinical trials of symptomatic ambulatory patients with COVID-19 with unclear benefit and potential trend to increased harm; therefore, it is not recommended for use outside of a clinical trial.

Complications of COVID-19 illness should be treated as they arise. Hospitalized patients with COVID-19 may be at increased risk for thromboembolic events. Several professional societies have issued guidelines regarding antithrombotic therapy in COVID-19. Generally, therapeutic anticoagulation should be considered in nonpregnant, hospitalized patients requiring supplemental oxygen but not intensive care if they have an elevated D-dimer and no serious bleeding risk. The risk of an adverse event from bleeding outweighs the potential benefit in critically ill patients. Pharmacologic venous thromboembolism prophylaxis should be considered for all other hospitalized patients unless there is a contraindication.

Medications such as angiotensin-converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARB) therapy should be continued if needed for concomitant medical conditions but not started as treatment for COVID-19. There is no evidence that use of nonsteroidal anti-inflammatory drugs (NSAIDs) is linked to worse outcomes, and either acetaminophen or NSAIDs can be used during the treatment of COVID-19.

Respiratory management of the nonintubated and intubated COVID-19 patient should take into consideration the tendency toward hypoxia. Nonpharmacologic adjunctive measures such as frequent repositioning and ambulation may be helpful.

Treatment references

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Post–COVID-19 Infection

Viral PCR tests in patients may remain positive for at least 3 months regardless of symptoms. However, even patients with lingering symptoms are generally not considered infectious, as virus is rarely if ever able to be cultured from the upper respiratory tract of patients after 10 days of illness.

Although infection with coronaviruses may confer some degree of immunity to reinfection, the duration and effectiveness of immunity following COVID-19 remain difficult to quantify and depend upon multiple host and viral factors. Patients who have had COVID-19, including people with prolonged post-

COVID-19 symptoms, are still recommended to receive all eligible vaccinations to reduce the risk of reinfection. They can do so when clinically recovered from infection and completing the isolation period. Neutralizing antibodies are detected in most patients following SARS-COV-2 infection, but the levels of these are more variable than in people after vaccinations. These likely provide protection against clinically apparent reinfection in most immunocompetent people for at least 3 and up to 6 months, but this time frame may be shorter if a new antigenically distinct variant emerges. Symptoms associated with reinfection tend to be similar to or milder than initial infections.

Post–COVID-19 infection reference

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Prevention of COVID-19

COVID-19 vaccination

For a detailed discussion of the vaccines approved in the United States, see COVID-19 vaccine.

Vaccination is the most effective way to prevent severe illness and death from COVID-19, including from the Delta and Omicron variants. In the United States in the fall of 2021, unvaccinated people were much more likely to die from COVID-19 than vaccinated people (with booster) (1).

In the United States, the CDC recommends immunocompetent people ages 12 years and older receive 1 dose of the updated 2023–2024 SARS-CoV-2 (COVID-19) mRNA vaccine from either manufacturer (Moderna or Pfizer-BioNTech) or 1 or 2 doses (depending on vaccination history) of the updated 2023-2024 SARS-CoV-2 virus (COVID-19) recombinant spike protein nanoparticle vaccine produced by Novavax (see CDC: Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States). Children ages 6 months to 4 years should receive 1 to 3 doses of an mRNA 2023–2024 formulation (recommended doses and manufacturer vary depending on vaccination history), and children 5 to 11 years should receive 1 dose regardless of vaccination history of either mRNA 2023–2024 formulation. The CDC also recommends adults ages 65 years and over receive an additional updated 2023-2024 COVID-19 vaccine dose (see CDC: Older Adults Now Able to Receive Additional Dose of Updated COVID-19 Vaccine). Guidance for people who are moderately or severely immunocompromised can also be found at the CDC website. The updated 2023–2024 vaccines target the XBB.1.5 Omicron variant. The 2023–2024 formulation replaces the COVID-19 monovalent vaccines that targeted the original SARS-CoV-2 strain and the bivalent vaccines that targeted both the original virus and BA.4/BA.5 Omicron subvariants.

Multiple COVID-19 vaccines are currently in use worldwide. For more information on global vaccine approvals and clinical trials, see the World Health Organization's COVID-19 vaccine tracker and landscape.

mRNA vaccines do not contain viral antigen but rather deliver a small, synthetic piece of mRNA that encodes for the desired target antigen (the spike protein). After being taken up by cells of the immune system, the vaccine mRNA degrades after instructing the cell to produce viral antigen. The antigen is then released and triggers the desired immune response to prevent severe infection upon subsequent exposure to the actual virus.

Adenovirus vector vaccines contain a piece of the DNA, or genetic material, that is used to make the distinctive "spike" protein of the SARS-CoV-2 virus, which then triggers the desired immune response.

Protein subunit adjuvanted vaccines contain a recombinant SARS-COV-2 spike protein along with an adjuvant that triggers the desired immune response. This is a classic vaccine approach that has been used in the United States for over 30 years.

Two mRNA vaccines (2023–2024 formulations of Moderna COVID-19 Vaccine and Pfizer-BioNTech COVID-19 Vaccine), and one protein subunit adjuvanted vaccine (2023–2024 formulation of Novavax COVID-19 Vaccine, Adjuvanted) are used in the United States (see CDC: COVID-19 Vaccination Clinical & Professional Resources). As of May 6, 2023, the adenovirus vector vaccine (Ad26.COV2.S) is no longer available for use in the United States due to the risk of serious adverse events. There is a plausible causal relationship between the adenovirus vector vaccine and a rare and serious adverse event—blood clots with low platelets (vaccine-induced thrombosis with thrombocytopenia syndrome, or VITTS).

Monoclonal antibody

Pemivibart, a monoclonal antibody, has received emergency use authorization (EUA) to prevent COVID-19 in people ages 12 and older and weighing at least 40 kg. It is for people who are not currently infected with SARS-CoV-2, who have not had a known recent exposure to an individual infected with SARS-CoV-2, and who have moderate-to-severe immune compromise and are unlikely to mount an adequate immune response to COVID-19 vaccination. Patients should still receive their COVID-19 vaccine. Pemivibart can be given at least 2 weeks after receiving a COVID-19 vaccine as an additional layer of protection.

Exposure prevention

In addition to staying up to date with COVID-19 vaccines, people can avoid being exposed to the virus by washing hands frequently, wearing face masks, maintaining social distance, avoiding poorly ventilated spaces and crowds, and taking other steps recommended by the Centers for Disease Control and Prevention (CDC). People should also have COVID-19 testing if they are exposed to an infected individual or have symptoms. Symptomatic individuals should follow recommendations for isolation.

To help prevent spread of SARS-CoV-2 from suspected cases, health care practitioners should use standard, contact, and airborne precautions with eye protection (2). Airborne precautions are particularly relevant for patients undergoing aerosol-generating procedures.

Prevention references

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More Information

The following English-language resources may be useful. Please note that THE MANUAL is not responsible for the content of these resources.

CDC: COVID Data Tracker
CDC: COVID-19 Vaccination Clinical & Professional Resources
CDC: Healthcare Workers: Information on COVID-19
WHO: Coronavirus disease (COVID-19) technical guidance
WHO: Novel Coronavirus (COVID-2019) situation reports
Infectious Diseases Society of America (IDSA) Guidelines on the Treatment and Management of Patients with COVID-19

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